

■ 시나리오 별 백업 복구(User Managed)

KEY WORDS

- <시나리오 9> TX 진행중 undo datafile 손상(백업 이후 리두정보 X)
- <시나리오 10> Temp file 손상
- <시나리오 11> All Data files, Control files, Redologfiles 있는 디스크 손상

<<시나리오 9>> TX 진행중 undo datafile 손상(백업 이후 리두정보 X)

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a, v$tablespace b
where a.ts# = b.ts#;
```

FILE#	TBS_NAME	FILE_NAME	CHECKPOINT_CHANGE#	STATUS
3	SYSAUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	2315457	ONLINE
1	SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf	2315457	SYSTEM
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	2315457	ONLINE
7	USERS	/u01/app/oracle/oradata/ORCL/users01.dbf	2315457	ONLINE

```
SYS@orcl> save tscheck.sql
```

```
SYS@orcl> sho parameter undo_tablespace
```

```
undo_tablespace          string          UNDOTBS1
```

```
SYS@orcl> select segment_id, segment_name, owner, tablespace_name, status from dba_rollback_segs;
```

SEGMENT_ID	SEGMENT_NAME	OWNER	TABLESPACE_NAME	STATUS
0	SYSTEM	SYS	SYSTEM	ONLINE
1	_SYSSMU1_1261223759\$	PUBLIC	UNDOTBS1	ONLINE
2	_SYSSMU2_27624015\$	PUBLIC	UNDOTBS1	ONLINE
3	_SYSSMU3_2421748942\$	PUBLIC	UNDOTBS1	ONLINE
4	_SYSSMU4_625702278\$	PUBLIC	UNDOTBS1	ONLINE
5	_SYSSMU5_2101348960\$	PUBLIC	UNDOTBS1	ONLINE
6	_SYSSMU6_813816332\$	PUBLIC	UNDOTBS1	ONLINE
7	_SYSSMU7_2329891355\$	PUBLIC	UNDOTBS1	ONLINE
8	_SYSSMU8_399776867\$	PUBLIC	UNDOTBS1	ONLINE
9	_SYSSMU9_1692468413\$	PUBLIC	UNDOTBS1	ONLINE
10	_SYSSMU10_930580995\$	PUBLIC	UNDOTBS1	ONLINE

11 rows selected.

```
SYS@orcl> select * from v$log;
```

GROUP#	THREAD#	SEQUENCE#	BYTES	BLOCKSIZE	MEMBERS	ARC	STATUS	FIRST_CHANGE#
--------	---------	-----------	-------	-----------	---------	-----	--------	---------------

```

FIRST_TIME      NEXT_CHANGE#
-----
NEXT_TIME      CON_ID
-----
          1          1          100  10485760          512          1 NO  INACTIVE          2304882  05-
MAY-25          2310234
06-MAY-25          0

          2          1          101  10485760          512          1 NO  INACTIVE          2310234  06-
MAY-25          2315457
06-MAY-25          0

          3          1          102  10485760          512          1 NO  CURRENT          2315457  06-
MAY-25          1.8447E+19
          0

```

SYS@orcl> save log.sql

SYS@orcl> select name dbname, open_mode, log_mode, checkpoint_change# from v\$database;

```

DBNAME  OPEN_MODE      LOG_MODE      CHECKPOINT_CHANGE#
-----
ORCL    READ WRITE      NOARCHIVELOG      2315457

```

SYS@orcl> save db.sql

SYS@ora19c> select s.username, s.sid, s.serial#, t.xidusn undo_seg_num, r.name undo_seg_name, t.ubafil undo_datafile_num, t.ubablk undo_block, t.used_ublk from v\$session s, v\$transaction t, v\$rollname r where s.taddr = t.addr and t.xidusn = r.usn;

no rows selected

SYS@orcl> save txroll.sql

<<hr>>

HR@orcl> select salary from hr.employees where employee_id = 102;

```

SALARY
-----
17000

```

HR@orcl> update hr.employees set salary = salary * 1.1 where employee_id = 102;

1 row updated.

HR@orcl> select salary from hr.employees where employee_id = 102;

```

SALARY
-----
18700

```

<<sys>>

SYS@ora19c> @txroll

USERNAME	SID	SERIAL#	UNDO_SEG_NUM	UNDO_SEG_NAME	UNDO_DATAFILE_NUM
UNDO_BLOCK	USED_UBLK				
HR	36	32073	9	_SYSSMU9_2100413762\$	2
1					468

SYS@orcl> @log

GROUP#	THREAD#	SEQUENCE#	BYTES	BLOCKSIZE	MEMBERS	ARC	STATUS	FIRST_CHANGE#	
FIRST_TIME	NEXT_CHANGE#								
1	1	100	10485760	512	1	NO	INACTIVE	2304882	05-
MAY-25	2310234								
06-MAY-25	0								
2	1	101	10485760	512	1	NO	INACTIVE	2310234	06-
MAY-25	2315457								
06-MAY-25	0								
3	1	102	10485760	512	1	NO	CURRENT	2315457	06-
MAY-25	1.8447E+19								
	0								

■ 장애 발생

SYS@orcl> ! ls -l /u01/app/oracle/oradata/ORCL/undotbs01.dbf

-rw-r-----. 1 oracle dba 361766912 May 6 09:41 /u01/app/oracle/oradata/ORCL/undotbs01.dbf

SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/undotbs01.dbf

SYS@orcl> ! ls -l /u01/app/oracle/oradata/ORCL/undotbs01.dbf

ls: cannot access /u01/app/oracle/oradata/ORCL/undotbs01.dbf: No such file or directory

■ 데이터베이스 비정상 종료

SYS@ora19c> alter system checkpoint;

ERROR:

*ERROR at line 1:

ORA-03113: end-of-file on communication channel

Process ID: 2220

Session ID: 220 Serial number: 43673

-- 오류나지 않을 경우 shutdown abort

SYS@ora19c> conn / as sysdba

Connected to an idle instance.

-- 여기서 idle instance 오류나지 않을 경우 shutdown abort

SYS@orcl> startup

ORACLE instance started.

Total System Global Area 1140849904 bytes

Fixed Size 8895728 bytes

Variable Size 754974720 bytes

Database Buffers 369098752 bytes

Redo Buffers 7880704 bytes

Database mounted.

ORA-01157: cannot identify/lock data file 4 - see DBWR trace file

ORA-01110: data file 4: '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'

SYS@orcl> select * from v\$recover_file;

FILE#	ONLINE	ONLINE_ ERROR	CHANGE#	TIME
CON_ID				
4	ONLINE	ONLINE	FILE NOT FOUND	0
0				

SYS@orcl> @tscheck

FILE#	TBS_NAME	FILE_NAME	CHECKPOINT_CHANGE#	STATUS
1	SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf	2319969	SYSTEM
3	SYS_AUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	2319969	ONLINE
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	2319969	ONLINE
7	USERS	/u01/app/oracle/oradata/ORCL/users01.dbf	2319969	ONLINE

SYS@orcl> select status from v\$instance;

STATUS

MOUNTED

- Recovery - 문제가 발생한 undo datafile을 offline drop 수행

SYS@orcl> alter database datafile 4 offline drop;

-- 또는 SYS@ora19c> alter database datafile '/u01/app/oracle/oradata/ORCL/undotbs01.dbf' offline drop;

Database altered.

```
SYS@orcl> alter database datafile 4 offline drop;
```

Database altered.

```
SYS@orcl> @tscheck
```

FILE#	TBS_NAME	FILE_NAME	CHECKPOINT_CHANGE#	STATUS
1	SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf	2319969	SYSTEM
3	SYS_AUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	2319969	ONLINE
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	2319969	RECOVER
7	USERS	/u01/app/oracle/oradata/ORCL/users01.dbf	2319969	ONLINE

-- 데이터베이스 open 수행

```
SYS@orcl> alter database open;
```

Database altered.

<<hr>>

```
[oracle@oel7v9r1 ~]$ sqlplus hr/hr
```

```
HR@orcl> update hr.employees set salary = 2000 where employee_id = 100;
```

1 row updated.

<<sys>>

-- hr에서 업데이트시 오류가 발생할 수 있지만 당시의 시점에 따라 랜덤으로 시스템언두를 사용함으로 오류가 발생하지 않도록 오라클이 자체적으로 제어할 수 있다.(사용자가 관여할 수 없다.)

```
SYS@orcl> @txroll
```

USERNAME	SID	SERIAL#
HR	62	42998

UNDO_SEG_NUM	UNDO_SEG_NAME	UNDO_DATAFILE_NUM	UNDO_BLOCK	USED_UBLK
0	SYSTEM	1	530	1

```
SYS@ora19c> select segment_id, segment_name, owner, tablespace_name, status from dba_rollback_segs;
```

SEGMENT_ID	SEGMENT_NAME	OWNER	TABLESPACE_NAME	STATUS
0	SYSTEM	SYS	SYSTEM	ONLINE
1	_SYSSMU1_1230671320\$	PUBLIC	UNDOTBS1	NEEDS RECOVERY
2	_SYSSMU2_3151764486\$	PUBLIC	UNDOTBS1	NEEDS RECOVERY
3	_SYSSMU3_1295854473\$	PUBLIC	UNDOTBS1	NEEDS RECOVERY
4	_SYSSMU4_1947518621\$	PUBLIC	UNDOTBS1	NEEDS RECOVERY
5	_SYSSMU5_1526312829\$	PUBLIC	UNDOTBS1	NEEDS RECOVERY
6	_SYSSMU6_2680029761\$	PUBLIC	UNDOTBS1	NEEDS RECOVERY
7	_SYSSMU7_2189229235\$	PUBLIC	UNDOTBS1	NEEDS RECOVERY
8	_SYSSMU8_4078897598\$	PUBLIC	UNDOTBS1	NEEDS RECOVERY
9	_SYSSMU9_2100413762\$	PUBLIC	UNDOTBS1	NEEDS RECOVERY
10	_SYSSMU10_2605481461\$	PUBLIC	UNDOTBS1	NEEDS RECOVERY

11 rows selected.

<<hr>>

HR@orcl> rollback;

Rollback complete.

<<sys>>

SYS@orcl> @txroll

no rows selected

- 새로운 undo tablespace 생성

SYS@orcl> create undo tablespace undo1 datafile '/u01/app/oracle/oradata/ORCL/undo01.dbf' size 100m autoextend on;

Tablespace created.

SYS@orcl> alter system set undo_tablespace = undo1;

System altered.

SYS@orcl> select segment_id, segment_name, owner, tablespace_name, status from dba_rollback_segs;

SEGMENT_ID	SEGMENT_NAME	OWNER	TABLESPACE_NAME	STATUS
0	SYSTEM	SYS	SYSTEM	ONLINE
1	_SYSSMU1_1261223759\$	PUBLIC	UNDOTBS1	NEEDS RECOVERY
2	_SYSSMU2_27624015\$	PUBLIC	UNDOTBS1	NEEDS RECOVERY

3 _SYSSMU3_2421748942\$	PUBLIC UNDOTBS1	NEEDS RECOVERY
4 _SYSSMU4_625702278\$	PUBLIC UNDOTBS1	NEEDS RECOVERY
5 _SYSSMU5_2101348960\$	PUBLIC UNDOTBS1	NEEDS RECOVERY
6 _SYSSMU6_813816332\$	PUBLIC UNDOTBS1	NEEDS RECOVERY
7 _SYSSMU7_2329891355\$	PUBLIC UNDOTBS1	NEEDS RECOVERY
8 _SYSSMU8_399776867\$	PUBLIC UNDOTBS1	NEEDS RECOVERY
9 _SYSSMU9_1692468413\$	PUBLIC UNDOTBS1	NEEDS RECOVERY
10 _SYSSMU10_930580995\$	PUBLIC UNDOTBS1	NEEDS RECOVERY

SEGMENT_ID	SEGMENT_NAME	OWNER	TABLESPACE_NAME	STATUS
11	_SYSSMU11_2468800258\$	PUBLIC	UNDO1	ONLINE
12	_SYSSMU12_891451708\$	PUBLIC	UNDO1	ONLINE
13	_SYSSMU13_523321169\$	PUBLIC	UNDO1	ONLINE
14	_SYSSMU14_856587081\$	PUBLIC	UNDO1	ONLINE
15	_SYSSMU15_744512766\$	PUBLIC	UNDO1	ONLINE
16	_SYSSMU16_2875759616\$	PUBLIC	UNDO1	ONLINE
17	_SYSSMU17_113360085\$	PUBLIC	UNDO1	ONLINE
18	_SYSSMU18_3760675133\$	PUBLIC	UNDO1	ONLINE
19	_SYSSMU19_3678045251\$	PUBLIC	UNDO1	ONLINE
20	_SYSSMU20_861131726\$	PUBLIC	UNDO1	ONLINE

21 rows selected.

<<hr>>

HR@orcl> update hr.employees set salary = 2000 where employee_id = 100;

1 row updated.

<<sys>>

SYS@orcl> @txroll

USERNAME	SID	SERIAL#		

UNDO_SEG_NUM	UNDO_SEG_NAME	UNDO_DATAFILE_NUM	UNDO_BLOCK	USED_UBLK

HR				
	18 _SYSSMU18_3760675133\$	5	229	1

- 트랜잭션을 hr에서 걸고 확인해보면 새로운 undo1 Ts사용
- 만약 새로운 undo1를 사용하지 않고 system을 사용한다면 문제가 될 수 있으므로 db를 내렸다 올리면 해결
- 해결되지 않는다면 문제가 됨으로 csr을 올려봐야 할것(오라클 소프트웨어가 문제)

<<hr>>

```
HR@orcl> rollback;
Rollback complete.
```

```
<<sys>>
```

```
-- 정보 확인
```

```
SYS@orcl> @tscheck
```

FILE#	TBS_NAME	FILE_NAME	CHECKPOINT_CHANGE#	STATUS
3	SYSAUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	2419974	ONLINE
1	SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf	2419974	SYSTEM
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	2319969	RECOVER
7	USERS	/u01/app/oracle/oradata/ORCL/users01.dbf	2419974	ONLINE
5	UNDO1	/u01/app/oracle/oradata/ORCL/undo01.dbf	2421354	ONLINE

- 기존 테이블스페이스를 제거하려고 시도

```
SYS@orcl> drop tablespace undotbs1 including contents and datafiles;
```

```
drop tablespace undotbs1 including contents and datafiles
```

```
*
```

```
ERROR at line 1:
```

```
ORA-01548: active rollback segment '_SYSSMU1_1261223759$' found, terminate dropping tablespace
```

- 기존 사용하고 있던 트랜잭션이 걸려있는 테이블스페이스는 삭제되지 않아 오류가 발생
- 수동으로 needs recovery가 되어 있는 테이블스페이스를 offline으로 수동 변환

```
SYS@orcl> select segment_name||',' from dba_rollback_segs where status = 'NEEDS RECOVERY';
```

```
SEGMENT_NAME||','
```

```
-----
_SYSSMU1_1261223759$,
_SYSSMU2_27624015$,
_SYSSMU3_2421748942$,
_SYSSMU4_625702278$,
_SYSSMU5_2101348960$,
_SYSSMU6_813816332$,
_SYSSMU7_2329891355$,
_SYSSMU8_399776867$,
_SYSSMU9_1692468413$,
_SYSSMU10_930580995$,
```

```
10 rows selected.
```

■ Recovery

SYS@ora19c> create pfile from spfile;

File created.

SYS@orcl> shutdown immediate

-- 수동으로 undo segment를 offline으로 수행하는 작업(정상종료 필요)

SYS@orcl> !vi \$ORACLE_HOME/dbs/initorcl.ora → 아래 내용을 추가

```
_offline_rollback_segments=(
_SYSSMU1_1261223759$,
_SYSSMU2_27624015$,
_SYSSMU3_2421748942$,
_SYSSMU4_625702278$,
_SYSSMU5_2101348960$,
_SYSSMU6_813816332$,
_SYSSMU7_2329891355$,
_SYSSMU8_399776867$,
_SYSSMU9_1692468413$,
_SYSSMU10_930580995$)
```

SYS@orcl> startup pfile=\$ORACLE_HOME/dbs/initorcl.ora
ORACLE instance started.

Total System Global Area 1140849904 bytes
Fixed Size 8895728 bytes
Variable Size 754974720 bytes
Database Buffers 369098752 bytes
Redo Buffers 7880704 bytes
Database mounted.
Database opened.

SYS@orcl> show parameter spfile

NAME	TYPE	VALUE

spfile	string	

SYS@orcl> drop tablespace undotbs1 including contents and datafiles;

Tablespace dropped.

- 정보 확인

SYS@orcl> @tscheck

FILE#	TBS_NAME	FILE_NAME	CHECKPOINT_CHANGE#	STATUS
-------	----------	-----------	--------------------	--------

```

-----
3 SYSAUX          /u01/app/oracle/oradata/ORCL/sysaux01.dbf      2422449 ONLINE
1 SYSTEM          /u01/app/oracle/oradata/ORCL/system01.dbf  2422449 SYSTEM
7 USERS    /u01/app/oracle/oradata/ORCL/users01.dbf      2422449 ONLINE
5 UNDO1          /u01/app/oracle/oradata/ORCL/undo01.dbf  2422449 ONLINE

```

SYS@orcl> select segment_id, segment_name, owner, tablespace_name, status from dba_rollback_segs;

```

SEGMENT_ID SEGMENT_NAME          OWNER  TABLESPACE_NAME STATUS
-----
0 SYSTEM          SYS    SYSTEM          ONLINE
11 _SYSSMU11_2468800258$ PUBLIC UNDO1          ONLINE
12 _SYSSMU12_891451708$ PUBLIC UNDO1          ONLINE
13 _SYSSMU13_523321169$ PUBLIC UNDO1          ONLINE
14 _SYSSMU14_856587081$ PUBLIC UNDO1          ONLINE
15 _SYSSMU15_744512766$ PUBLIC UNDO1          ONLINE
16 _SYSSMU16_2875759616$ PUBLIC UNDO1          ONLINE
17 _SYSSMU17_113360085$ PUBLIC UNDO1          ONLINE
18 _SYSSMU18_3760675133$ PUBLIC UNDO1          ONLINE
19 _SYSSMU19_3678045251$ PUBLIC UNDO1          ONLINE
20 _SYSSMU20_861131726$ PUBLIC UNDO1          ONLINE

```

11 rows selected.

<원래대로 돌리기 1>

SYS@orcl> shut immediate

Database closed.

Database dismounted.

ORACLE instance shut down.

SYS@orcl> startup → spfile로 시작하기

ORACLE instance started.

Total System Global Area 1140849904 bytes

Fixed Size 8895728 bytes

Variable Size 754974720 bytes

Database Buffers 369098752 bytes

Redo Buffers 7880704 bytes

Database mounted.

Database opened.

SYS@orcl> show parameter undo_tablespace

```

NAME                                TYPE      VALUE
-----
undo_tablespace                      string    UNDO1

```

SYS@orcl> show parameter spfile

NAME	TYPE	VALUE
spfile	string	/u01/app/oracle/product/19.3.0 /dbhome_1/dbs/spfileorcl.ora

SYS@orcl> alter system set undo_tablespace = undotbs1 scope=spfile;

System altered.

SYS@orcl> shut abort

ORACLE instance shut down.

SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/*.*

SYS@orcl> ! cp -v /home/oracle/backup/noarch/* /u01/app/oracle/oradata/ORCL/
 '/home/oracle/backup/noarch/control01.ctl' -> '/u01/app/oracle/oradata/ORCL/control01.ctl'
 '/home/oracle/backup/noarch/control02.ctl' -> '/u01/app/oracle/oradata/ORCL/control02.ctl'
 '/home/oracle/backup/noarch/initiorcl.ora' -> '/u01/app/oracle/oradata/ORCL/initiorcl.ora'
 '/home/oracle/backup/noarch/redo01.log' -> '/u01/app/oracle/oradata/ORCL/redo01.log'
 '/home/oracle/backup/noarch/redo02.log' -> '/u01/app/oracle/oradata/ORCL/redo02.log'
 '/home/oracle/backup/noarch/redo03.log' -> '/u01/app/oracle/oradata/ORCL/redo03.log'
 '/home/oracle/backup/noarch/sysaux01.dbf' -> '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'
 '/home/oracle/backup/noarch/system01.dbf' -> '/u01/app/oracle/oradata/ORCL/system01.dbf'
 '/home/oracle/backup/noarch/temp01.dbf' -> '/u01/app/oracle/oradata/ORCL/temp01.dbf'
 '/home/oracle/backup/noarch/undotbs01.dbf' -> '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'
 '/home/oracle/backup/noarch/users01.dbf' -> '/u01/app/oracle/oradata/ORCL/users01.dbf'

SYS@orcl> startup

ORACLE instance started.

Total System Global Area 1140849904 bytes

Fixed Size 8895728 bytes

Variable Size 754974720 bytes

Database Buffers 369098752 bytes

Redo Buffers 7880704 bytes

Database mounted.

Database opened.

SYS@orcl> show parameter undo_tablespace

NAME	TYPE	VALUE
undo_tablespace	string	UNDOTBS1

SYS@orcl> select segment_id, segment_name, owner, tablespace_name, status from dba_rollback_segs;

SEGMENT_ID	SEGMENT_NAME	OWNER	TABLESPACE_NAME	STATUS
------------	--------------	-------	-----------------	--------

```

-----
0 SYSTEM          SYS      SYSTEM      ONLINE
1 _SYSSMU1_1261223759$ PUBLIC UNDOTBS1 ONLINE
2 _SYSSMU2_27624015$ PUBLIC UNDOTBS1 ONLINE
3 _SYSSMU3_2421748942$ PUBLIC UNDOTBS1 ONLINE
4 _SYSSMU4_625702278$ PUBLIC UNDOTBS1 ONLINE
5 _SYSSMU5_2101348960$ PUBLIC UNDOTBS1 ONLINE
6 _SYSSMU6_813816332$ PUBLIC UNDOTBS1 ONLINE
7 _SYSSMU7_2329891355$ PUBLIC UNDOTBS1 ONLINE
8 _SYSSMU8_399776867$ PUBLIC UNDOTBS1 ONLINE
9 _SYSSMU9_1692468413$ PUBLIC UNDOTBS1 ONLINE
10 _SYSSMU10_930580995$ PUBLIC UNDOTBS1 ONLINE

```

11 rows selected.

<원래대로 돌리기 2>

SYS@orcl> create pfile from spfile;

File created.

SYS@orcl> shutdown abort

ORACLE instance shut down.

SYS@orcl> ! cp -v backup/noarch/* /u01/app/oracle/oradata/ORCL/

'backup/noarch/control01.ctl' -> '/u01/app/oracle/oradata/ORCL/control01.ctl'

'backup/noarch/itorcl.ora' -> '/u01/app/oracle/oradata/ORCL/itorcl.ora'

'backup/noarch/redo01.log' -> '/u01/app/oracle/oradata/ORCL/redo01.log'

'backup/noarch/redo02.log' -> '/u01/app/oracle/oradata/ORCL/redo02.log'

'backup/noarch/redo03.log' -> '/u01/app/oracle/oradata/ORCL/redo03.log'

'backup/noarch/sysaux01.dbf' -> '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'

'backup/noarch/system01.dbf' -> '/u01/app/oracle/oradata/ORCL/system01.dbf'

'backup/noarch/temp01.dbf' -> '/u01/app/oracle/oradata/ORCL/temp01.dbf'

'backup/noarch/undotbs01.dbf' -> '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'

'backup/noarch/users01.dbf' -> '/u01/app/oracle/oradata/ORCL/users01.dbf'

SYS@orcl> !vi \$ORACLE_HOME/dbs/itorcl.ora

-- *.undo_tablespace='UNDO1' -> *.undo_tablespace='UNDOTBS1' 변경 필요

SYS@orcl> startup pfile=\$ORACLE_HOME/dbs/itorcl.ora

ORACLE instance started.

Database mounted.

Database opened.

SYS@orcl> create spfile from pfile;

File created.

```
[oracle@oel7v9r1 ~]$ cd /u01/app/oracle/oradata/ORCL/
[oracle@oel7v9r1 ~]$ ls
control01.ctl  redo01.log  redo03.log  system01.dbf  undo01.dbf  users01.dbf
initorcl.ora   redo02.log  sysaux01.dbf  temp01.dbf    undotbs01.dbf
[oracle@oel7v9r1 ~]$ rm -rf undo01.dbf
```

```
SYS@orcl> select segment_id, segment_name, owner, tablespace_name, status from dba_rollback_segs;
```

SEGMENT_ID	SEGMENT_NAME	OWNER	TABLESPACE_NAME	STATUS
0	SYSTEM	SYS	SYSTEM	ONLINE
1	_SYSSMU1_1261223759\$	PUBLIC	UNDOTBS1	ONLINE
2	_SYSSMU2_27624015\$	PUBLIC	UNDOTBS1	ONLINE
3	_SYSSMU3_2421748942\$	PUBLIC	UNDOTBS1	ONLINE
4	_SYSSMU4_625702278\$	PUBLIC	UNDOTBS1	ONLINE
5	_SYSSMU5_2101348960\$	PUBLIC	UNDOTBS1	ONLINE
6	_SYSSMU6_813816332\$	PUBLIC	UNDOTBS1	ONLINE
7	_SYSSMU7_2329891355\$	PUBLIC	UNDOTBS1	ONLINE
8	_SYSSMU8_399776867\$	PUBLIC	UNDOTBS1	ONLINE
9	_SYSSMU9_1692468413\$	PUBLIC	UNDOTBS1	ONLINE
10	_SYSSMU10_930580995\$	PUBLIC	UNDOTBS1	ONLINE

11 rows selected.

- 기존 생성했던 undo01.dbf 파일이 그대로 있음으로 해당 파일 삭제
- 있어도 상관은 없으나 새로 생성시 동일 이름 생성 불가

<<시나리오 10>> Temp file 손상

```
SYS@orcl> select name from v$tempfile;
```

```
NAME
```

```
-----  
/u01/app/oracle/oradata/ORCL/temp01.dbf
```

```
SYS@orcl> select tablespace_name, contents from dba_tablespaces;
```

```
TABLESPACE_NAME CONTENTS
```

```
-----  
SYSTEM          PERMANENT  
SYSAUX           PERMANENT  
UNDOTBS1         UNDO  
TEMP             TEMPORARY  
USERS            PERMANENT
```

```
SYS@orcl> select username, temporary_tablespace from dba_users;
```

```
USERNAME
```

```
-----  
TEMPORARY_TABLESPACE
```

```
-----  
SYS  
TEMP
```

```
SYSTEM  
TEMP
```

```
37 rows selected.
```

```
SYS@orcl> select property_value from database_properties where property_name='DEFAULT_TEMP_TABLESPACE';
```

```
PROPERTY_VALUE
```

```
-----  
TEMP
```

```
SYS@orcl> show parameter sort_area_size
```

```
NAME                                TYPE      VALUE  
-----  
sort_area_size                      integer    65536
```

■ 장애 발생

```
SYS@orcl> ! ls -l /u01/app/oracle/oradata/ORCL/temp01.dbf
-rw-r-----. 1 oracle dba 33562624 May  6 10:30 /u01/app/oracle/oradata/ORCL/temp01.dbf
```

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/temp01.dbf
```

```
SYS@orcl> ! ls -l /u01/app/oracle/oradata/ORCL/temp01.dbf
ls: cannot access /u01/app/oracle/oradata/ORCL/temp01.dbf: No such file or directory
```

<<hr>>

```
HR@orcl> select s.*, b.* from all_objects s, all_objects b order by 1,2,3,4;
select s.*, b.* from all_objects s, all_objects b order by 1,2,3,4
*
```

```
ERROR at line 1:
ORA-01116: error in opening database file 201
ORA-01110: data file 201: '/u01/app/oracle/oradata/ORCL/temp01.dbf'
ORA-27041: unable to open file
Linux-x86_64 Error: 2: No such file or directory
Additional information: 3
```

■ Recovery

<<sys>>

- tempfile 추가

```
SYS@orcl> alter tablespace temp add tempfile '/u01/app/oracle/oradata/ORCL/temp02.dbf' size 10m autoextend on;
```

Tablespace altered.

```
SYS@orcl> select name from v$tempfile;
```

NAME

/u01/app/oracle/oradata/ORCL/temp01.dbf
/u01/app/oracle/oradata/ORCL/temp02.dbf

- 손상된 temp file 삭제

```
SYS@orcl> alter tablespace temp drop tempfile '/u01/app/oracle/oradata/ORCL/temp01.dbf';
Tablespace altered.
```

```
SYS@orcl> select name from v$tempfile;
```

NAME

/u01/app/oracle/oradata/ORCL/temp02.dbf

- temp file은 정상종료 후 시작하면 oracle이 자동으로 생성.

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/temp02.dbf
```

```
SYS@orcl> shut immediate
```

```
SYS@orcl> startup
```

```
SYS@orcl> ! ls -l /u01/app/oracle/oradata/ORCL/temp02.dbf
```

```
-rw-r-----. 1 oracle dba 10493952 May  6 10:46 /u01/app/oracle/oradata/ORCL/temp02.dbf
```

```
SYS@orcl> select name from v$tempfile;
```

```
NAME
```

```
-----  
-----
```

```
/u01/app/oracle/oradata/ORCL/temp02.dbf
```

```
SYS@orcl> SYS@orcl> select * from dba_temp_files;
```

FILE_NAME	FILE_ID	TABLESPACE_NAME	BYTES	BLOCKS
STATUS	RELATIVE_FNO	AUT	MAXBYTES	MAXBLOCKS

INCREMENT_BY	USER_BYTES	USER_BLOCKS	SHARED	INST_ID

/u01/app/oracle/oradata/ORCL/temp02.dbf	2	TEMP	10485760	1280 ONLINE
2 YES 3.4360E+10	4194302			
1	9437184	1152	SHARED	

- temp file 추가

```
SYS@orcl> alter tablespace temp add tempfile '/u01/app/oracle/oradata/ORCL/temp01.dbf' size 10m;  
Tablespace altered.
```

- temp file의 크기 조정

```
SYS@orcl> alter database tempfile '/u01/app/oracle/oradata/ORCL/temp01.dbf' resize 50m;  
Database altered.
```

- temp file의 자동 확장기능 활성화

```
SYS@orcl> alter database tempfile '/u01/app/oracle/oradata/ORCL/temp01.dbf' autoextend on;  
Database altered.
```

- temp file 삭제

```
SYS@orcl> alter tablespace temp drop tempfile '/u01/app/oracle/oradata/ORCL/temp02.dbf';
alter tablespace temp drop tempfile '/u01/app/oracle/oradata/ORA19C/temp02.dbf'
```

*

ERROR at line 1:

ORA-25152: TEMPFILE cannot be dropped at this time

-- 지금 사용중이라 삭제 불가시

```
SYS@orcl> select b.tablespace, b.segfile#, b.segblk#, b.blocks, a.username, a.sid, a.serial# from v$session a, v$sort_usage b
where a.saddr = b.session_addr;
```

TABLESPACE	SEGFILE#	SEGBLK#	BLOCKS	USERNAME	SID	SERIAL#
TEMP	202	256	128	SYS	237	50419

```
SYS@orcl> alter system kill session '237,50419' immediate;
```

System altered.

```
SYS@ora19c> alter tablespace temp drop tempfile '/u01/app/oracle/oradata/ORA19C/temp02.dbf';
```

Tablespace altered.

- 자동확장기능 비활성화

```
SYS@orcl> alter database tempfile '/u01/app/oracle/oradata/ORCL/temp01.dbf' autoextend off;
```

-- 생성시 너무 작게는 안된다.(extent 크기 이상)

```
SYS@orcl> alter tablespace temp add tempfile '/u01/app/oracle/oradata/ORCL/temp02.dbf' size 1m;
alter tablespace temp add tempfile '/u01/app/oracle/oradata/ORCL/temp02.dbf' size 1m
```

*

ERROR at line 1:

ORA-03214: File Size specified is smaller than minimum required

■ Temp Tablespace 추가 변경 예제

```
SYS@orcl> select * from dba_temp_files;
```

FILE_NAME	FILE_ID	TABLESPACE_NAME	BYTES	BLOCKS
STATUS	RELATIVE_FNO	AUT	MAXBYTES	MAXBLOCKS

INCREMENT_BY	USER_BYTES	USER_BLOCKS	SHARED	INST_ID
--------------	------------	-------------	--------	---------

```

-----
/u01/app/oracle/oradata/ORCL/temp01.dbf          1 TEMP          52428800      6400 ONLINE
1 NO              0    0
              0   51380224      6272 SHARED

```

- 새로운 temp tablespace를 생성한 후 default temporary tablespace로 지정

```

SYS@orcl> create temporary tablespace newtemp tempfile '/u01/app/oracle/oradata/ORCL/newtemp01.dbf' size 10m
autoextend on;

```

Tablespace created.

```

SYS@orcl> alter database default temporary tablespace newtemp;

```

Database altered.

```

SYS@orcl> select property_value from database_properties where property_name='DEFAULT_TEMP_TABLESPACE';

```

```

PROPERTY_VALUE
-----

```

```

NEWTEMP

```

- 기존 temp 테이블스페이스 삭제

```

SYS@orcl> drop tablespace temp including contents and datafiles;
Tablespace dropped.

```

- 원본 복구

```

SYS@orcl> create temporary tablespace temp tempfile '/u01/app/oracle/oradata/ORCL/temp01.dbf' size 10m autoextend on;

```

Tablespace created.

```

SYS@orcl> alter database default temporary tablespace temp;

```

Database altered.

```

SYS@orcl> drop tablespace newtemp including contents and datafiles;

```

Tablespace dropped.

```

SYS@orcl> select name from v$tempfile;

```

```

NAME
-----

```

```

/u01/app/oracle/oradata/ORCL/temp01.dbf

```

<<시나리오11>> 모든 데이터파일, 리두로그파일, 컨트롤파일이 있는 디스크 손상, 복구

■ 백업 본 체크

```
SYS@orcl> show parameter spfile;
```

NAME	TYPE	VALUE

spfile	string	/u01/app/oracle/product/19.3.0 /dbhome_1/dbs/spfileorcl.ora

```
SYS@orcl> create pfile from spfile;
```

File created.

```
SYS@orcl> select name from v$datafile
```

```
2 union
3 select name from v$controlfile
4 union
5 select member from v$logfile
6 union
7 select name from v$tempfile
```

NAME

```
-----
/u01/app/oracle/oradata/ORCL/control01.ctl
/u01/app/oracle/oradata/ORCL/redo01.log
/u01/app/oracle/oradata/ORCL/redo02.log
/u01/app/oracle/oradata/ORCL/redo03.log
/u01/app/oracle/oradata/ORCL/sysaux01.dbf
/u01/app/oracle/oradata/ORCL/system01.dbf
/u01/app/oracle/oradata/ORCL/temp01.dbf
/u01/app/oracle/oradata/ORCL/undotbs01.dbf
/u01/app/oracle/oradata/ORCL/users01.dbf
```

9 rows selected.

```
SYS@orcl> save db.sql
```

```
SYS@orcl> show parameter control_files
```

NAME	TYPE	VALUE

control_files	string	/u01/app/oracle/oradata/ORCL /control01.ctl

- 컨트롤 파일 위치 변경(pfile 이용)

```
SYS@orcl> !mkdir -p /home/oracle/oradata
```

SYS@orcl> !vi \$ORACLE_HOME/dbs/initiorcl.ora

*.control_files='/home/oracle/oradata/control01.ctl' → 경로 수정

- 데이터파일, 리두로그파일의 위치 변경

SYS@orcl> select 'alter database rename file '''||name||''''' to '''/home/oracle/oradata/''';' from v\$datafile
union all

select 'alter database rename file '''||name||''''' to '''/home/oracle/oradata/''';' from v\$tempfile
union all

select 'alter database rename file '''||member||''''' to '''/home/oracle/oradata/''';' from v\$logfile;

2 3 4 5

'ALTERDATABASERENAMEFILE'''||NAME||'''''TO'''/HOME/ORACLE/ORADATA/''';'

alter databse rename file '/u01/app/oracle/oradata/ORCL/system01.dbf' to '/home/oracle/oradata/system01.dbf;'
alter databse rename file '/u01/app/oracle/oradata/ORCL/undotbs01.dbf' to '/home/oracle/oradata/undotbs01.dbf;'
alter databse rename file '/u01/app/oracle/oradata/ORCL/sysaux01.dbf' to '/home/oracle/oradata/sysaux01.dbf;'
alter databse rename file '/u01/app/oracle/oradata/ORCL/users01.dbf' to '/home/oracle/oradata/users01.dbf;'
alter databse rename file '/u01/app/oracle/oradata/ORCL/temp01.dbf' to '/home/oracle/oradata/temp01.dbf;'
alter databse rename file '/u01/app/oracle/oradata/ORCL/redo05.log' to '/home/oracle/oradata/redo01.log;'
alter databse rename file '/u01/app/oracle/oradata/ORCL/redo06.log' to '/home/oracle/oradata/redo02.log;'
alter databse rename file '/u01/app/oracle/oradata/ORCL/redo07.log' to '/home/oracle/oradata/redo03.log;'

■ 장애 발생

SYS@orcl> !rm /u01/app/oracle/oradata/ORCL/*

SYS@orcl> alter system checkpoint;

alter system checkpoint

*

ERROR at line 1:

ORA-03113: end-of-file on communication channel

Process ID: 16406

Session ID: 33 Serial number: 18880

SYS@orcl> conn /as sysdba

Connected to an idle instance.

SYS@orcl> startup

ORACLE instance started.

Total System Global Area 1140849904 bytes

Fixed Size 8895728 bytes

Variable Size 754974720 bytes

Database Buffers 369098752 bytes

Redo Buffers 7880704 bytes

ORA-00205: error in identifying control file, check alert log for more info

```
SYS@orcl> select status from v$instance;
STATUS
-----
STARTED
```

```
SYS@orcl> ! ls -l /u01/app/oracle/oradata/ORCL/
total 0
```

■ RECOVERY - 새로운 위치로 복구.

- 새로운 위치로 백업본을 restore한다.

```
SYS@ora19c> shutdown abort
ORACLE instance shut down.
```

```
SYS@orcl> shut abort
ORACLE instance shut down.
```

```
SYS@orcl> !cp -v /home/oracle/backup/noarch/* /home/oracle/oradata/
'/home/oracle/backup/noarch/control01.ctl' -> '/home/oracle/oradata/control01.ctl'
'/home/oracle/backup/noarch/control02.ctl' -> '/home/oracle/oradata/control02.ctl'
'/home/oracle/backup/noarch/initorcl.ora' -> '/home/oracle/oradata/initorcl.ora'
'/home/oracle/backup/noarch/redo01.log' -> '/home/oracle/oradata/redo01.log'
'/home/oracle/backup/noarch/redo02.log' -> '/home/oracle/oradata/redo02.log'
'/home/oracle/backup/noarch/redo03.log' -> '/home/oracle/oradata/redo03.log'
'/home/oracle/backup/noarch/sysaux01.dbf' -> '/home/oracle/oradata/sysaux01.dbf'
'/home/oracle/backup/noarch/system01.dbf' -> '/home/oracle/oradata/system01.dbf'
'/home/oracle/backup/noarch/temp01.dbf' -> '/home/oracle/oradata/temp01.dbf'
'/home/oracle/backup/noarch/undotbs01.dbf' -> '/home/oracle/oradata/undotbs01.dbf'
'/home/oracle/backup/noarch/users01.dbf' -> '/home/oracle/oradata/users01.dbf'
```

```
[oracle@oel7v9r1 ~]$ vi $ORACLE_HOME/dbs/initorcl.ora
```

```
*.control_files='/home/oracle/oradata/control01.ctl' → 이전에 이미 완료.
```

- pfile을 이용해서 데이터베이스를 mount까지.

```
SYS@orcl> startup pfile=$ORACLE_HOME/dbs/initorcl.ora mount
ORACLE instance started.
```

```
Total System Global Area 1140849904 bytes
Fixed Size                  8895728 bytes
Variable Size               754974720 bytes
Database Buffers           369098752 bytes
Redo Buffers                7880704 bytes
Database mounted.
```

```
SYS@ora19c> select status from v$instance;
```

```
STATUS
```

```
-----
```

```
MOUNTED
```

```
SYS@orcl> @db
```

```
NAME
```

```
-----
```

```
/home/oracle/oradata/control01.ctl          ---> 컨트롤 파일 수정.
```

```
/u01/app/oracle/oradata/ORCL/redo01.log
```

```
/u01/app/oracle/oradata/ORCL/redo02.log
```

```
/u01/app/oracle/oradata/ORCL/redo03.log
```

```
/u01/app/oracle/oradata/ORCL/sysaux01.dbf
```

```
/u01/app/oracle/oradata/ORCL/system01.dbf
```

```
/u01/app/oracle/oradata/ORCL/temp01.dbf
```

```
/u01/app/oracle/oradata/ORCL/undotbs01.dbf
```

```
/u01/app/oracle/oradata/ORCL/users01.dbf
```

```
9 rows selected.
```

- 데이터파일, 리두로그 파일, Temp 파일을 새로운위치로 변경

```
alter database rename file '/u01/app/oracle/oradata/ORCL/system01.dbf' to '/home/oracle/oradata/system01.dbf';
```

```
alter database rename file '/u01/app/oracle/oradata/ORCL/undotbs01.dbf' to '/home/oracle/oradata/undotbs01.dbf';
```

```
alter database rename file '/u01/app/oracle/oradata/ORCL/sysaux01.dbf' to '/home/oracle/oradata/sysaux01.dbf';
```

```
alter database rename file '/u01/app/oracle/oradata/ORCL/users01.dbf' to '/home/oracle/oradata/users01.dbf';
```

```
alter database rename file '/u01/app/oracle/oradata/ORCL/temp01.dbf' to '/home/oracle/oradata/temp01.dbf';
```

```
alter database rename file '/u01/app/oracle/oradata/ORCL/redo01.log' to '/home/oracle/oradata/redo01.log';
```

```
alter database rename file '/u01/app/oracle/oradata/ORCL/redo02.log' to '/home/oracle/oradata/redo02.log';
```

```
alter database rename file '/u01/app/oracle/oradata/ORCL/redo03.log' to '/home/oracle/oradata/redo03.log';
```

```
SYS@orcl> alter database open;
```

```
Database altered.
```

```
SYS@orcl> @db
```

```
NAME
```

```
-----
```

```
-----
```

```
/home/oracle/oradata/control01.ctl
```

```
/home/oracle/oradata/redo01.log
```

```
/home/oracle/oradata/redo02.log
```

```
/home/oracle/oradata/redo03.log
```

```
/home/oracle/oradata/sysaux01.dbf
/home/oracle/oradata/system01.dbf
/home/oracle/oradata/temp01.dbf
/home/oracle/oradata/undotbs01.dbf
/home/oracle/oradata/users01.dbf
```

9 rows selected.

- 원래대로 돌리기

```
SYS@orcl> create pfile from spfile;
```

```
SYS@orcl> shut immediate
```

```
SYS@orcl> !vi $ORACLE_HOME/dbs/initorcl.ora
```

```
*.control_files='/u01/app/oracle/oradata/ORCL/control01.ctl'          --> 수정
```

```
SYS@orcl> !cp -v /home/oracle/oradata/* /u01/app/oracle/oradata/ORCL/
'/home/oracle/oradata/control01.ctl' -> '/u01/app/oracle/oradata/ORCL/control01.ctl'
'/home/oracle/oradata/control02.ctl' -> '/u01/app/oracle/oradata/ORCL/control02.ctl'
'/home/oracle/oradata/initorcl.ora' -> '/u01/app/oracle/oradata/ORCL/initorcl.ora'
'/home/oracle/oradata/redo01.log' -> '/u01/app/oracle/oradata/ORCL/redo01.log'
'/home/oracle/oradata/redo02.log' -> '/u01/app/oracle/oradata/ORCL/redo02.log'
'/home/oracle/oradata/redo03.log' -> '/u01/app/oracle/oradata/ORCL/redo03.log'
'/home/oracle/oradata/sysaux01.dbf' -> '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'
'/home/oracle/oradata/system01.dbf' -> '/u01/app/oracle/oradata/ORCL/system01.dbf'
'/home/oracle/oradata/temp01.dbf' -> '/u01/app/oracle/oradata/ORCL/temp01.dbf'
'/home/oracle/oradata/undotbs01.dbf' -> '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'
'/home/oracle/oradata/users01.dbf' -> '/u01/app/oracle/oradata/ORCL/users01.dbf'
```

```
SYS@orcl>
```

```
alter database rename file '/home/oracle/oradata/system01.dbf' to '/u01/app/oracle/oradata/ORCL/system01.dbf';
alter database rename file '/home/oracle/oradata/undotbs01.dbf' to '/u01/app/oracle/oradata/ORCL/undotbs01.dbf';
alter database rename file '/home/oracle/oradata/sysaux01.dbf' to '/u01/app/oracle/oradata/ORCL/sysaux01.dbf';
alter database rename file '/home/oracle/oradata/users01.dbf' to '/u01/app/oracle/oradata/ORCL/users01.dbf';
alter database rename file '/home/oracle/oradata/temp01.dbf' to '/u01/app/oracle/oradata/ORCL/temp01.dbf';
alter database rename file '/home/oracle/oradata/redo01.log' to '/u01/app/oracle/oradata/ORCL/redo01.log';
alter database rename file '/home/oracle/oradata/redo02.log' to '/u01/app/oracle/oradata/ORCL/redo02.log';
alter database rename file '/home/oracle/oradata/redo03.log' to '/u01/app/oracle/oradata/ORCL/redo03.log';
```

```
SYS@orcl> alter database open;
```

```
SYS@orcl> @db
```

```
NAME
```

```
-----
-----
```

```
/u01/app/oracle/oradata/ORCL/temp01.dbf  
/u01/app/oracle/oradata/ORCL/control01.ctl  
/u01/app/oracle/oradata/ORCL/redo01.log  
/u01/app/oracle/oradata/ORCL/redo02.log  
/u01/app/oracle/oradata/ORCL/redo03.log  
/u01/app/oracle/oradata/ORCL/sysaux01.dbf  
/u01/app/oracle/oradata/ORCL/system01.dbf  
/u01/app/oracle/oradata/ORCL/undotbs01.dbf  
/u01/app/oracle/oradata/ORCL/users01.dbf
```

```
SYS@orcl> !rm /home/oracle/oradata/*.*
```